UNIT IV: Digital Twin, Simulation, AR/VR and Cybersecurity

Subject: 23A03703d - Smart Manufacturing

1. Digital Twin Technology and Virtual Prototyping

1.1 Recap of the digital twin

A digital twin is a living virtual model of a physical product, machine, line or plant. It is updated with real data, so it shows the current state and can predict future behaviour. (Unit II introduced the idea; this unit focuses on design, simulation and security around it.)

KEY: Digital twin = physical asset + virtual model + data link (ideally two-way).

1.2 Digital twin across the product life cycle

- Design: the twin is the CAD and simulation model used to test ideas.
- Build: the twin helps plan the factory layout and test PLC code virtually (virtual commissioning).
- Operate: live data keeps the twin in step with the real asset.
- Maintain: the twin predicts failures and plans service.
- Retire or upgrade: the twin records history, helping redesign the next product.

1.3 Virtual prototyping

Virtual prototyping builds and tests a product on the computer before any physical prototype is made.

- Benefits: fewer physical prototypes, faster design cycles, earlier detection of faults, lower material cost.
- Methods: finite element analysis (FEA) for stress, computational fluid dynamics (CFD) for airflow and cooling, multibody dynamics for motion, thermal analysis for heat.

- Example: a car engine part is tested for stress, temperature and fatigue in simulation before the first casting is poured.

1.4 Virtual commissioning

Virtual commissioning connects the real PLC program to a simulated plant. Errors in logic, interlocks and sequences are found while the machine is still on paper, reducing start-up time on the shop floor.

1.5 Digital twin benefits and difficulties

- Benefits: better decisions, reduced risk, faster time-to-market, knowledge retained in the model.
- Difficulties: building accurate models is expensive; models must be updated; data integration across vendors is hard; skills in simulation and data are scarce.

Exam Q: Explain how virtual commissioning reduces start-up time.

2. Simulation Software for Smart Manufacturing

2.1 Why simulate?

Simulation lets engineers test a system under many conditions without disrupting real production. It answers questions such as: how many machines do we need? where is the bottleneck? what happens if demand doubles?

2.2 Types of simulation

- Discrete event simulation (DES): models events such as job arrivals and machine completions. Best for production lines, queues and logistics.
- Continuous simulation: models quantities that change smoothly, such as temperature, level or chemical concentration.
- Agent-based simulation: models many autonomous agents (machines, AGVs, operators) and their interactions.
- Physics-based simulation: FEA, CFD and multibody models for product and process behaviour.

- Monte Carlo simulation: uses random sampling to study uncertainty, for example variation in cycle times.

2.3 Common simulation tools (examples)

- Plant and logistics: Siemens Plant Simulation, FlexSim, AnyLogic, Arena.
- Physics and engineering: ANSYS, COMSOL, Abaqus, OpenFOAM (CFD).
- Robot simulation: RoboDK, Visual Components, Process Simulate.
- Process simulation: Aspen Plus and HYSYS for chemical plants.
- Open-source options: Python libraries such as SimPy for discrete event simulation.

2.4 Steps in a simulation study

1. Define the objective and the questions to answer.
2. Collect input data (cycle times, failure rates, arrival patterns).
3. Build the model at the right level of detail.
4. Verify the model (does it do what we intended?).
5. Validate the model (does it match real behaviour?).
6. Run experiments and compare scenarios.
7. Analyse results and communicate recommendations.

KEY: A simulation is only as good as its data and its validation; a fancy animation proves nothing on its own.

2.5 Example: line balancing

A line has five stations with cycle times 40, 55, 52, 38 and 60 seconds. A discrete event simulation shows that station 5 limits the output to one unit every 60 seconds. Moving one task from station 5 to station 4 balances the line and raises output without new machines.

Exam Q: What is the difference between verification and validation of a simulation model?

3. Augmented Reality (AR) and Virtual Reality (VR) in Indu

3.1 Definitions

- Virtual reality (VR): a fully computer-generated environment. The user wears a headset and is immersed in a virtual world.
- Augmented reality (AR): digital information is overlaid on the real world, usually through a headset, tablet or smartphone camera.
- Mixed reality (MR): real and virtual objects interact in real time; the virtual objects respond to the real environment.

3.2 AR and VR in manufacturing

- Training: operators practise machine setup or safety procedures in VR before touching real equipment.
- Maintenance: AR glasses show step-by-step repair instructions and wiring diagrams over the machine.
- Remote assistance: an expert sees what the technician sees and draws annotations on the live view.
- Assembly guidance: AR shows which bolt to tighten next and torque values, reducing errors.
- Design review: VR walk-throughs of a factory layout before construction.
- Quality inspection: AR overlays tolerance limits on the part during inspection.

3.3 Hardware and software

- VR headsets: fully immersive displays with tracking of head and hand movement.
- AR devices: smart glasses, tablets, head-mounted displays with cameras and sensors.
- Software: 3D engines such as Unity and Unreal Engine; CAD integration; tracking libraries; IoT data feeds.

3.4 Benefits and challenges

- Benefits: lower training cost, fewer errors, faster repairs, safer practice, knowledge capture from experts.

- Challenges: hardware cost, comfort and eye strain, need for accurate 3D models, network needs for remote help, and the time needed to create content.

KEY: AR and VR are most valuable where the cost of a mistake on the real machine is high.

3.5 Example

A technician fixing a compressor sees the wiring path highlighted on the machine through AR glasses. A remote expert places an arrow on the live view to show a valve. The repair takes 40 minutes instead of two hours spent searching manuals.

Exam Q: Give one example each of AR and VR use in a factory.

4. Cybersecurity in Smart Manufacturing: Threats and Ris

4.1 Why security matters now

Traditional factories were isolated: machines were not connected to the internet. Smart factories connect OT and IT, which opens many new paths for attackers. A single infected laptop can spread malware to controllers that have no protection.

4.2 Types of threats

- Malware and ransomware: encrypt files or stop production until a ransom is paid.
- Phishing and social engineering: staff tricked into giving passwords or opening malicious files.
- Insider threats: employees or contractors who misuse access, intentionally or by mistake.
- Denial of service (DoS): flooding networks so that controllers cannot communicate.
- Man-in-the-middle attacks: intercepting and changing messages between devices.

- Unauthorised access: default passwords on PLCs and HMI panels left unchanged.
- Supply chain attacks: malicious code inserted into software or firmware updates.
- Physical attacks: tampering with USB ports or sensors on the shop floor.

4.3 Risks in smart manufacturing

- Safety risk: manipulated machine settings can injure people or damage equipment.
- Production risk: downtime, lost orders and missed deliveries.
- Quality risk: subtle changes to process data can produce defective products without alarms.
- Intellectual property risk: theft of designs and recipes.
- Financial and legal risk: penalties, insurance claims, regulatory breaches.
- Reputational risk: loss of customer trust.

4.4 Why OT security is different from IT security

Aspect	IT systems	OT systems
Priority	Confidentiality	Availability and safety first
Patching	Frequent and quick	Limited by downtime and certification
Lifetime	3 to 5 years	15 to 30 years
Protocols	Standard internet protocols	Old industrial protocols with no security
Response	Shut down and reboot	Shutdown may be dangerous

KEY: In OT, the rule is "safety, then availability, then integrity, then confidentiality".

4.5 Attack example

Stuxnet (discovered in 2010) is a well-known industrial malware that targeted PLCs. It showed that attackers can reach control systems through IT networks and USB devices. Lessons: segment networks, control removable media and monitor controller behaviour.

Exam Q: Why is patching industrial systems harder than patching office computers?

5. Cybersecurity Frameworks for Industrial Systems

5.1 Purpose of a framework

A framework gives a structured way to identify assets, assess risks, select controls, respond to incidents and improve over time.

5.2 IEC 62443

- International series of standards for security of industrial automation and control systems (IACS).
- Introduces security levels (SL 1 to SL 4) and the concept of zones and conduits: groups of assets with similar risk, and the controlled communication paths between them.
- Covers the whole life cycle: product developers, system integrators and asset owners each have requirements.

5.3 NIST Cybersecurity Framework (CSF)

- Developed by the US National Institute of Standards and Technology. Version 2.0 (2024) adds the govern function.
- Six core functions: Govern, Identify, Protect, Detect, Respond and Recover.
- Widely used because it is flexible and technology-neutral; it can be applied to plants and suppliers.

5.4 ISO/IEC 27001 and 27019

- ISO/IEC 27001 defines an information security management system (ISMS).

- ISO/IEC 27019 gives guidance for the energy sector control systems.

5.5 Defence in depth

Security is built in layers so that a single failure does not expose everything:

- Physical security: locked cabinets, controlled access to the shop floor.
- Network segmentation: firewalls and DMZ between OT and IT.
- Host security: antivirus or allow-listing on engineering workstations.
- Application security: secure coding and signed updates.
- Monitoring: intrusion detection for industrial protocols.
- People and process: training, incident response plans, backups.

KEY: No framework replaces a risk assessment; the framework tells you what to check, and the risk assessment tells you what matters most.

Exam Q: What are the six core functions of the NIST CSF?

6. Best Practices for Securing IoT and IIoT Devices

6.1 Secure design and procurement

- Choose devices with secure boot, signed firmware and encrypted communication.
- Ask suppliers for a software bill of materials (SBOM).
- Avoid devices with hard-coded default passwords.
- Require support and security update commitments for the device lifetime.

6.2 Identity and access

- Give each device a unique identity and certificate.
- Use strong, unique passwords and change defaults on installation.
- Apply the principle of least privilege: each user and device gets only what it needs.
- Use multi-factor authentication for remote access.

6.3 Network security

- Segment networks into zones; limit traffic between zones.
- Disable unused ports and services.
- Use VPNs or secure gateways for remote access; never expose PLCs directly to the internet.
- Monitor network traffic and watch for unusual industrial protocol commands.

6.4 Data protection

- Encrypt data in transit with TLS and, where possible, OPC-UA security.
- Protect data at rest with encryption and access control.
- Keep backups offline or immutable, and test restores regularly.

6.5 Maintenance and monitoring

- Maintain an inventory of all connected devices.
- Patch on a tested schedule; use compensating controls when patching is not possible.
- Log events centrally and review logs regularly.
- Run vulnerability scans and penetration tests with care to avoid disrupting production.

6.6 People and response

- Train operators and engineers to recognise phishing and unusual machine behaviour.
- Prepare an incident response plan with clear roles and communication steps.
- Practise drills so that the team knows what to shut down, what to isolate and whom to call.
- Share threat intelligence with industry groups and computer emergency response teams.

6.7 Checklist for an IIoT deployment

- Inventory of devices and their owners.
- Risk assessment and zone design.
- Secure configuration and identity for each device.

- Encrypted communication with certificates.
- Monitoring, logging and alerting in place.
- Backup, recovery and response plan tested.

Exam Q: List three best practices for securing IIoT sensors.

1.6 Digital twin architecture for manufacturing (ISO 2324

ISO 23247 is an international framework for digital twins in manufacturing. It describes four main entities:

- Observable manufacturing elements (OMEs): the real things such as machines, tools, materials and people.
- Data collection and device control: sensors and controllers that gather and act on data.
- Digital twin entity: the virtual model with its data, logic and visualisation.
- User application: the people and systems that use the twin, for example planners or maintenance teams.

KEY: Keep the framework in mind: real elements feed data into a twin, and users act on the twin's results.

2.6 Simulation and digital twins together

- A simulation is often a one-time study; a digital twin keeps running with live data.
- Simulation models can be used inside a twin as its "what-if" engine.
- Example: a twin of a cooling system runs a physics model every minute with live temperature data; the operator tests a fan-speed change in the twin before applying it to the real plant.

4.6 Security tools commonly used in OT environments

- Firewalls designed for industrial protocols (Modbus, OPC-UA, PROFINET) that filter by function code.
- Industrial intrusion detection systems that learn normal communication patterns.
- Asset inventory tools that discover devices on the network and list firmware versions.
- Application allow-listing on engineering workstations so that only approved programs run.
- Data diodes: one-way hardware links that let data leave the OT network but not enter it.
- Secure remote access gateways with session recording.

5.6 Case study: securing a bottling plant

A bottling plant connected its PLCs to a cloud analytics platform to reduce waste. Early reviews found several issues: default passwords on HMI panels, a shared engineering laptop used for internet browsing, and open ports on the PLC network.

- Actions taken: passwords changed and unique accounts created; laptop separated into an engineering-only machine; firewall rules allowed only the required OPC-UA traffic; a data diode was added for data flowing to the cloud.
- Results: analytics continued without loss of function, and the audit found no unauthorised access in the following year.
- Lesson: most early wins come from basic hygiene, not from expensive tools.

6.8 Summary table of best practices

Area	Key practice	Example
Design	Secure by default	Disable unused services
Identity	Unique credentials	Per-user accounts on HMI
Network	Segmentation	Firewall between OT and IT
Data	Encryption	TLS on MQTT and OPC-UA
Operations	Patch and monitor	Scheduled maintenance window
People	Training and drills	Phishing awareness course

7. Worked Examples and Revision

7.1 Worked example: zone design for a small plant

- Zone A: PLCs and drives on the production line. Access only from the engineering workstation.
- Zone B: SCADA and HMI servers. Allowed to talk to Zone A through a firewall with allow-listed Modbus or OPC-UA traffic.
- Zone C: MES and office IT network. Connected to Zone B through a data diode or a tightly controlled gateway.
- Remote vendors connect through a VPN into a jump server in a separate DMZ, with recorded sessions.

7.2 Worked example: responding to ransomware on an HMI

1. Detect: monitoring alerts on unusual file encryption.
2. Contain: isolate the infected HMI from the network; keep the line in a safe state.
3. Eradicate: rebuild the HMI from a known-good image; check PLC programs against backups.
4. Recover: restore data, test functions, then restart production in stages.

5. Learn: update the zone rules and training.

7.3 Revision checklist

- Can you explain how digital twins support virtual commissioning?
- Can you name three simulation types and their uses?
- Can you distinguish AR, VR and MR?
- Can you list four threats to a smart factory?
- Can you explain why OT security puts safety first?
- Can you name the six NIST CSF functions?
- Can you describe zones and conduits in IEC 62443?

7.4 Short answer practice

Exam Q: Differentiate between verification and validation in simulation.

Exam Q: Why should PLCs never be directly exposed to the internet?

Exam Q: Explain defence in depth with three layers.